

The 2026–2027 Season Plan

RANGER class · MATE ROV Competition

Eleven months. Mid-June 2026 to Dauphin Island.

Second revision — June 2026. The competition-grade build and the administration track, now with the student-owned trade-study framing, the January Q&A track, the weight budget, the registration window, and an honest contingency. Phase 1 has its own companion plan.

Strategy

A new RANGER team will not out-build experienced teams in ten months. But it can out-prepare them on documentation, presentation, and safety — together, forty percent of the total score. So this plan front-loads skills, fundamentals, and the company's identity over the summer and fall, and treats the spring as execution against a known manual. Everything is in service of one deliverable, late April 2027: Dauphin Island.

The build commitment stands: a competition-grade, reliable platform — brushless thrusters, a professionally terminated tether, a sealed enclosure, an aluminum frame — chosen less for raw performance than for reliability and reuse. A vehicle that does not leak and does not fail mid-run protects the demonstration score, and every major component carries into future seasons. One thing about that commitment is sharpened in this revision: the budget commits the money, but the students commit the design. Each major architecture choice runs through a real trade study in Phase 3, and the students confirm or amend the funded candidates and own the rationale. When a judge asks why six thrusters, the answer must trace to the team's decision matrix — not to this document. That is both the spirit and the letter of Section 4.3.

The budget is honest: roughly \$7,800 to build, plus travel — the increase over the prior revision is a real contingency on the long-lead order, not new hardware.

The plan

Phase one — Foundation. Mid-June through July 2026.

Six weeks. The goal is not a competition ROV. It is a team that exists, a working starter ROV in the water, and kids who have soldered something. Form the company — CEO, CFO, mechanical lead, electrical lead, software lead, communications lead. Build the SeaMATE Triggerfish kit as the team's first ROV, powered by a small 12 V sealed lead-acid battery the team supplies — the kit ships without a power source. Get every kid in a pool by week three. Lock pool access for the year, including one deep-water host. Hold a soldering bootcamp, an electronics workshop, and a project-notebook session, and set the safety culture on day one — every meeting opens with a safety brief. Add a waterproofing exercise: each kid designs a small sealed PVC capsule with a single cable penetration, drops it, recovers it, checks the paper towel inside, and iterates until it stays dry across three consecutive dives — the foundation skill for the float build in Phase 2. Start the community-outreach log on day one, and start the administrative work — registration, waivers, insurance, fundraising — in parallel.

The Foundation phase has its own companion document — the Phase 1 Plan — laying out the six meetings, the gate into Phase 2, and the parent track in detail.

SeaMATE Triggerfish ROV kit	\$950
12 V SLA battery, charger, enclosure, and Powerpole leads	\$125

Two soldering stations, irons, and consumables	\$170
Two multimeters and a hand-tool set	\$130
Waterproofing capsule exercise (PVC, end caps, O-rings, epoxy)	\$50
Heat-shrink, wire, zip ties, and build consumables	\$55
Ballast, trim weights, and flotation	\$20
Bench-test bin and transport totes	\$40
Safety gear (first aid kit, fire extinguisher, fume fan, mats)	\$100
Subtotal	\$1,640

Phase two — Capability. August through mid-October 2026.

Ten weeks. School starts. Settle into a weekly cadence: one weeknight, one weekend block. Each sub-team takes on a mini-project that maps to a real RANGER skill. Mechanical designs and 3D-prints a working gripper. Electrical builds a fused topside control box with proper terminations. Software writes a Python program that detects and counts objects in a webcam feed. The float team builds a v1 vertical profiler with a syringe buoyancy engine that hits 2.5 m and transmits depth — built on a NiMH battery, the chemistry the rules require, since alkaline and lithium-polymer are not permitted. Every member completes the Onshape learning path and captures a household object with photogrammetry. The prototypes are throwaway. The learning is the deliverable.

Two practices run beneath the mini-projects. The Triggerfish stays in service as the team's pilot trainer — winning pilots fly weekly from October onward, and stick time is the cheapest points on the board. And the technical-documentation and marketing-display skeletons stand up in August, even when there is little to put in them: by December they are populated, and by March the team is refining, not creating.

Float v1 components (microcontroller, MS5837 sensor, LoRa pair, syringe, NiMH battery)	\$260
3D-printer filament (community maker-space access assumed)	\$50
Gripper and payload prototype materials	\$60
Topside control prototype (Raspberry Pi 4, gamepad, breadboard)	\$120
USB camera for computer-vision work	\$30
Outreach materials and workshop consumables	\$125
Subtotal	\$645

Phase three — Design. Mid-October through mid-January 2027.

Twelve weeks. The 2027 manual drops around November. Within forty-eight hours of release, the team reads it together. Then a two-week decomposition: every task broken into requirements, scoring, and constraints. Trade studies follow — frame architecture, thruster count and type,

payload tools, camera choices, float comms protocol. Every decision earns a written rationale in the project notebook; those rationales become the technical documentation in March. CAD locks by mid-December. Long-lead parts ship before the holiday break — and this is the order that matters most, because Blue Robotics lead times stretch from February onward as every team orders at once.

This is where the competition vehicle is bought — and where the decisions must belong to the students. The budget below funds a set of default candidates chosen for reliability and reuse: six Blue Robotics T200 thrusters, a pre-terminated tether (faulty hand-made terminations are a top failure mode for new teams), a sealed Blue Robotics enclosure, an aluminum T-slot frame, and a regulated topside power supply. The trade studies confirm or amend each candidate, and the students write the case either way. Two constraints frame every study. Weight: under 15 kg in air earns the +10-point weight tier, and a T200-equipped aluminum frame lands at 13 to 15 — the team designs to the tier rather than discovering its weight in April. And fluid power: a pneumatic manipulator is a trade-study outcome, not a pre-decision — choosing it brings the fluid-power SID, the fluid-power quiz, and the regulator rules with it, while a mechanical claw drops all three.

The topside bench replica is built in December, not late January. It mirrors the ROV electronics, so software develops over the holiday break while the vehicle is still a CAD file — and it becomes the demonstration rig whenever the real vehicle is in pieces.

Six Blue Robotics T200 thrusters	\$1,506
Six Basic ESC R3 motor controllers	\$240
Blue Robotics 4-inch watertight enclosure assembly	\$250
Aluminum T-slot frame extrusion and hardware	\$250
Pre-terminated competition tether (Fathom-class)	\$350
Topside power supply (regulated, 48 V)	\$300
Raspberry Pi 5 (ROV) and Arduino Mega (sensor interface)	\$190
Two underwater cameras (USB, DIY housings)	\$100
WetLink penetrators, eight	\$105
Main fuse and ANL/MIDI fuse holder	\$20
Clamp ammeter (full-load-amp measurement)	\$70
Topside bench replica (December — parallel software development)	\$70
Float v2 refinement parts	\$80
Contingency buffer (~10 percent of the long-lead order)	\$350
Subtotal	\$3,881

Note on the thruster decision. Six T200 thrusters are the largest single line in the plan. The order is placed in December, but it is worth a deliberate go/no-go in the fall: confirmed if the funding is in hand, deferred to bilge-pump conversions (about a fifth of the cost) if it is not. Either path uses the same frame, geometry, and control architecture — only the motors and drivers differ. The contingency line is real: Pi-class boards are volatile in the memory shortage, and enclosure pricing runs above the line item often enough to plan for.

Phase four — Build. Mid-January through February 2027.

Six weeks. Construction. Tether, topside control box, ROV frame, electronics integration, float build, and the manipulator the trade study selected. Software lands on hardware it already knows from the December bench replica. Plan for two weeks of “it doesn't work” before it works. Bare frame wet by end of January. Electronics integrated by mid-February. First full-system test by end of February. The fluid-power quiz (if pneumatics were chosen) and safety pre-inspection tracking begin now — and so does the weekly mock Q&A: a mentor and one outside adult, every week from January to the regional, because unfamiliar faces are what the kids will face in April.

Pneumatic manipulator (cylinder, solenoid, regulator, tubing) — if selected	\$175
Spare ESCs and electronics backups	\$80
Rework consumables (epoxy, sealants, O-rings)	\$80
Mounting hardware, fasteners, brackets	\$80
Replacement-parts buffer (one major break)	\$150
Display-board base materials	\$60
Misc build consumables	\$50
Subtotal	\$675

Phase five — Test and document. March 2027.

Four weeks. Full mission rehearsals at the deep-water pool every weekend. Time each task. Identify what breaks; fix it. Lock the SIDs — electrical, fluid power if used, non-ROV device. Write the JSEA, focused on poolside operations rather than shop safety. Draft the technical documentation — student-written, with any AI assistance disclosed, as the rules require — and alongside it the Company Spec Sheet and the float's design document (DOC-004), both scored deliverables in their own right. The weekly mock Q&A that began in January continues, now as full mock engineering presentations — live and oral, no slide deck permitted. Prepare the marketing display, including the digital copy now required.

Critical spares (ESC, penetrator, sensor backups)	\$200
Marketing display printing (36 × 48) and mounting	\$130
Documentation printing and binding	\$50
Backup hardware (spare Raspberry Pi, sensors)	\$100
Subtotal	\$480

Phase six — Polish and compete. April 2027 through the regional.

Documentation submission deadlines hit in early April. Confirm DISL's exact dates. Print the marketing display, thirty-six by forty-eight inches. Five engineering-presentation rehearsals, minimum, on top of the weekly mock Q&A that began in January. Pack the trailer. Drive to

Dauphin Island, late April 2027. On registration: the regional window opens in mid-February and closes in early April — the team registers in February, the week it opens, and treats the close date as a cliff, because a missed window ends the season regardless of what the robot can do.

Regional registration fee	\$250
Final marketing-display reprint and handouts	\$60
Team shirts (eight at \$20)	\$160
Day-of miscellaneous	\$50
Subtotal	\$520

The parallel track — administration, registration, and funding

Three workstreams run alongside the build from day one. They belong to the parents, not the students, and a missed deadline on any of them can end the season as surely as a leak.

Registration and compliance. Register the company with MATE. The Northern Gulf Coast regional registration window opens in mid-February 2027 and closes in early April — register the week it opens. Collect liability waivers, photo-and-media releases, and a medical and emergency form for every student. Confirm the program carries liability insurance; the pool host will likely ask for proof. Check whether the team's affiliation requires youth-protection background checks for adult mentors.

The DISL relationship. Contact Rachel McDonald at the Dauphin Island Sea Lab early — to confirm the 2027 competition dates, the pool's depth and dimensions, and exactly which documents the regional requires.

Funding. The build costs roughly \$7,800, and travel to the regional adds roughly \$2,000 more. That money has to be raised. Start a sponsor list and a fundraising plan in the summer, not the spring — and use the Phase 1 showcase in late July as the first sponsor event.

Skills the team will leave with

By April, every kid will have hands-on with: soldering and basic electronics, fusing and full-load-amp measurement, waterproofing — O-rings, epoxy potting, penetrators — 3D CAD in Onshape, 3D printing, Python with either Arduino or Raspberry Pi, OpenCV for image recognition, photogrammetry, pressure and depth sensors, wireless RF, pneumatics if selected, piloting, safety culture and hazard analysis, basic project management, technical writing, and public speaking with live Q&A. Specialists will go deeper in their lanes.

Budget summary

Working totals based on 2026 vendor pricing for Blue Robotics thrusters and enclosures, a Fathom-class tether, and the SeaMATE kit, with a real contingency on the long-lead order. Excludes pool access (donated facility) and project notebooks (kids bring their own). Travel is estimated rather than excluded.

Phase one — Foundation	\$1,640
Phase two — Capability	\$645
Phase three — Design	\$3,881
Phase four — Build	\$675
Phase five — Test and document	\$480
Phase six — Polish and compete	\$520
Build total	\$7,841
Travel and lodging to the regional (estimate)	~\$2,000
All-in season cost	~\$9,800

The travel estimate assumes carpooling and shared lodging for the six-hour drive to Dauphin Island; late April is the front edge of beach season, so quotes are gathered in the fall, not the spring. If the team wins the regional and advances to the World Championship, budget a further \$3,000–\$5,000 — upside, not baseline.

Milestones

End of July 2026	Starter ROV in the water. Deep-pool access confirmed. Company formed. Registration paperwork and fundraising underway.
November 2026	2027 manual released. Full-team read-through within forty-eight hours.
Mid-December 2026	CAD locked. Long-lead parts — thrusters, enclosure, tether — ordered before the holiday break. Bench replica built over the break.
January 2027	Weekly mock Q&A begins — a mentor and one outside adult, every week to the regional.
Mid-February 2027	Regional registration opens. The team registers that week.
End of February 2027	First full-system pool test.
Early April 2027	Registration window closes. Technical-documentation submission deadline. Confirm both with DISL.
Late April 2027	Northern Gulf Coast Regional. Dauphin Island Sea Lab.

Risks, and how to defuse them

Two failure modes kill new RANGER teams. The first is not getting in the water enough. Fix this by locking pool access in June, not October — and by keeping the Triggerfish flying weekly once school starts. The second is under-investing in the engineering-and-communications track because the ROV isn't done yet. Fix this by starting the marketing display and the technical-documentation skeleton in August, even when there is little to put in them.

A third risk lives at the parent level: over-functioning. The rule for the year is — teach a skill once, then step back and let them fail. The kids learning to recover from failure is what makes the engineering-presentation Q&A go well in April. This revision extends the rule to the plan itself: the funded architecture is a set of default candidates, and the students' trade studies — not this document — make the decisions.

Two more risks carry over from the first revision. Administrative deadlines: a missed registration date or an unsigned waiver can end the season as surely as a leak — treat the parallel track as a real workstream with a named owner, and remember the registration window closes in early April. Supply lead times: Blue Robotics lead times stretch from February as every team orders at once — place the long-lead order before the holiday break, which is why CAD locks in mid-December and why the contingency line now exists.

Where to start

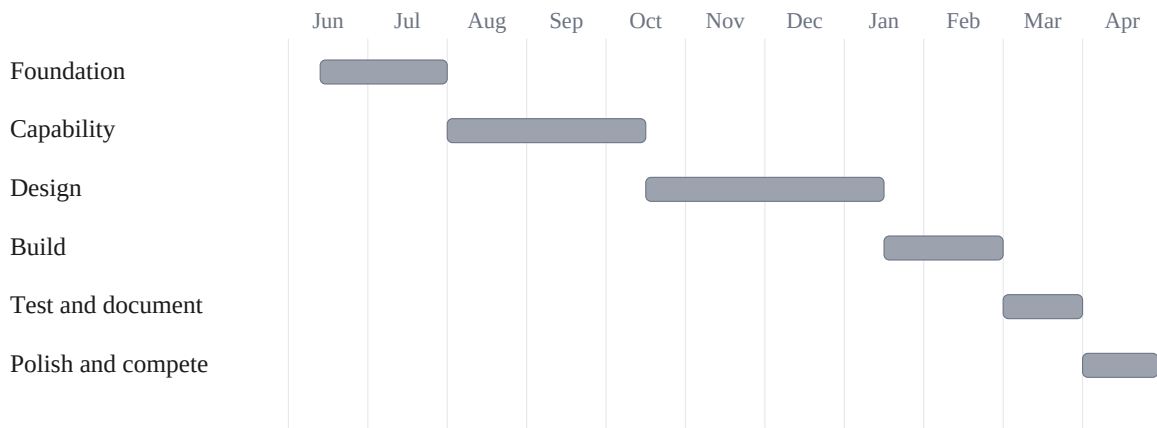
If we do nothing else in June: buy the SeaMATE Triggerfish kit, get six kids soldering, and put it in the water within three weeks. Start the outreach log, the waivers, and the sponsor conversations the same month. Everything else in this plan is downstream of that.

A note on the second season

The 2027 vehicle is built to be inherited. The aluminum frame, the sealed enclosure, the penetrators, the tether, the topside power supply, and the T200 thrusters all carry into the 2027-2028 build — the season's largest costs are one-time. A second season reuses the platform and iterates one subsystem, the pattern every multi-year winning program follows. Realistically, Season 2 runs closer to \$3,500–\$4,700 all-in against this season's roughly \$9,800 — the difference is simply not building the platform twice. The most important thing to protect for Season 2 is not hardware but people: cross-train two students per role, because graduation, not cost, is what ends young programs.

Appendix — Integrated Master Schedule

*One page. Six phases. Eleven months. The bars show duration; the milestones below mark the dates above.
Pin it to a wall.*



Milestones

End Jul	Starter ROV wet
Nov	2027 manual released
Mid-Dec	CAD locked, long-lead parts ordered
Jan	Weekly mock Q&A begins
Mid-Feb	Regional registration opens — register that week
End Feb	First full-system test
Early Apr	Registration closes; documentation due
Late Apr	Dauphin Island regional